

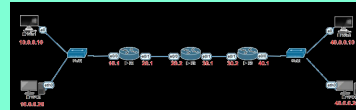
EIGRP

Enhanced interior gateway routing protocol
It is developed by cisco then its announced as open standard
It will send hello packet to form a neighborhood
Hello packet will send with multicast ip 224.0.0.10
It contains three table

- Neighbor table --- It contains neighbor information
- Topology table --- It contains best & backup route
- Routing table --- It contains only best route

Metric = Composite parameter or K-value

- K1 - Bandwidth
 - K2 - Load
 - K3 - Delay
 - K4 - Reliability
 - K5 - MTU
- Default k-value
- | | | | | |
|--------|--------|--------|--------|--------|
| K1 = 1 | K2 = 0 | K3 = 1 | K4 = 0 | K5 = 0 |
|--------|--------|--------|--------|--------|
- Metric formula
- $$= 256 \left(\frac{10^7}{\text{Least bandwidth}} + \text{Total delay} \right)$$



Neighborhood

By sending hello packet

HELLO Ext int IP 224.0.0.10 multicast IP

Neighborhood Conditions

Same N/w
same ASN
Same time

neighbor table
R1 - neighbor

Update packet
live -> neighbor

Timers

HELLO = 5 sec

Hold = 15 sec

Metric

It is use to calculate distance of a route

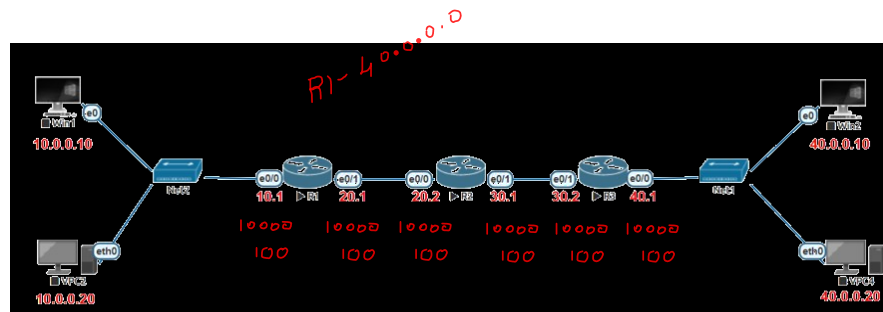
Composite parameter

K-value

Proc NET-M7

	K1 Bandwidth	K2 Load	K3 Delay	K4 K5 Reliability
256 x (	$K1 \times BW$	$\frac{K2 \times BW}{256 - LOAD}$	$K3 \times DLY$	$\frac{K5}{REL + K4}$
Default K Values:	K1 = 1 1 times Bandwidth is Bandwidth	K2 = 0 0 times anything is 0, and 0 divided by anything is 0	K3 = 1 1 times Delay is Delay	K4 = 0 K5 = 0 When K5 is 0, this section considered to result in 1
256 x (	BW	0	DLY	1

$K1$ ✓
 $K2$ ✗
 $K3$ ✓
 $K4$ ✗
 $K5$ ✗



$$= 256 \left\{ \frac{10^7}{\text{least bw}} + \text{Total Delay} \right\}$$

$$= 256 \left\{ \frac{10^7}{10,000} + 300 \right\}$$

$$256 \left\{ 1000 + 300 \right\}$$

$$256 \left\{ 1300 \right\}$$

$$= 332800$$